Seat No.:_____

AHU ID No._____ Name_____ Major_____ Grade_____

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Stony Brook Institute at Anhui University

2022 Fall Semester

Applied Calculus II Midterm Exam

(Closed-book Exam Duration: 120 minutes)

Section	I	II	III	IV	V	Total Score
Score						
Grader						

Score	

I. Blank Fillings (5 items, 2 points each)

1. Suppose $F(x)=\frac{2}{3}\ln\cos 2x$ is an antiderivative of $f(x)=k\tan 2x$, then $k=$_____.

2. If $\int f(x)dx=\sin x+C$, then $\int \frac{f(\arcsin x)}{\sqrt{1-x^2}}dx=$_____.

3. Calculate $\int_{-1}^{1} x|x|dx=$ _____.

4. If $f(x)=\int_a^x 12t^2dt$ and $\int_0^1 f(x)dx=1$, then $a=$_____.

5. If the improper integral $\int_0^{+\infty}\frac{dx}{1+kx^2}$ converges to 1 $(k>0)$, then $k=$_____.

Score	

II. Choices (5 items, 3 points each)

6. Suppose function $f(x)$ is continuous, $F(x)$ is an antiderivative of $f(x)$, then which one is correct? ( )

A. $F(x)$ is an even function. $\Leftrightarrow$ $f(x)$ is an odd function.

B. $F(x)$ is an odd function. $\Leftrightarrow$ $f(x)$ is an even function.

C. $F(x)$ is a periodic function. $\Leftrightarrow$ $f(x)$ is a periodic function.

D. $F(x)$ is a monotone function. $\Leftrightarrow$ $f(x)$ is a monotone function.

7. Let $M=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\left(\frac{\tan x}{1+x^4}+x^8\right)dx, N=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\left[\sin^8 x+\ln(x+\sqrt{x^2+1})\right]dx$,

$P=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\left(\tan^4 x+e^x\cos x-e^{-x}\cos x\right)dx$, then ( )

A. $P>N>M$.

B. $N>P>M$.

C. $N>M>P$.

D. $P>M>N$.

8. $\int_2^{8\pi+2}|\sin x|dx=$ ( )

A. $4\int_0^{2\pi}\sin xdx$.

B. $8\int_0^{\pi}\sin xdx$.

C. $4\int_0^{2\pi}\cos xdx$.

D. $8\int_0^{\pi}\cos xdx$.

9. If $\frac{d}{dx}\int_0^{e^{-x}}f(t)dt=e^x$,then $f(x)=$ ( )

A. $-e^{-2x}$.

B. e^{2x}.

C. $-x^{-2}$.

D. x^2.

10. Improper integrals $I_1=\int_0^1\frac{dx}{\sqrt{x}(1+x)}, I_2=\int_1^{+\infty}\frac{dx}{\sqrt{x}(1+x)}$,which one is correct? ( )

A. I_1 is convergent, I_2 is divergent.

B. I_1 is divergent, I_2 is convergent.

C. I_1 is divergent, I_2 is divergent.

D. I_1 is convergent, I_2 is convergent.

Score	

III. Calculations (7 items, 8 points each)

11. Calculate $\lim_{n\to\infty}\frac{1}{n}\left(\sin\frac{\pi}{n}+\sin\frac{2\pi}{n}+\cdots+\sin\frac{(n-1)\pi}{n}\right)$.

12. Compute $\int_1^{64} \frac{\sqrt[3]{x}}{x(\sqrt{x}+\sqrt[3]{x})}dx$.

13. Calculate $\int \sec^3 x dx$.

14. Evaluate $\int \frac{3}{1+x^3}dx$.

15. Compute $\lim_{x\to 0}\dfrac{\left(\int_0^x e^{t^2}\,dt\right)^2}{\int_0^x te^{t^2}\,dt}$.

16. Calculate $\int_{\frac{1}{2}}^{\frac{3}{2}}\dfrac{dx}{\sqrt{\left|x-x^2\right|}}$.

17. Suppose $f(x)$ is a continuous function and $f(x)=2x+2\int_0^1 f(t)dt$, find $f(x)$.

IV. Proofs (2 items, 7 points each)

Score	

18. Suppose $f(x)$ is continuous on $[a,b]$, prove that the equality holds

$$\int_a^b f(x)dx = \int_a^b f(a+b-x)dx.$$

19. If the improper integral $\int_1^{+\infty} f^2(x)dx$ converges, show that the improper integral $\int_1^{+\infty} \frac{f(x)}{x}dx$ absolutely converges.

Score	

V. Analytical calculation (5 points)

20. Use the mean value theorem of the definite integral to estimate the upper bound and lower bound of the following integral value

$$\int_0^{2\pi} \frac{1}{1+0.5\cos x}\,dx .$$